

1.

OPERATING SYSTEM

TYPES

Batch OS

Jobs are collected and executed in Batches.

Time Sharing OS

Many users interact with the system simultaneously.

RealTime OS

Responds to inputs within strict time constraints.

Distributed OS

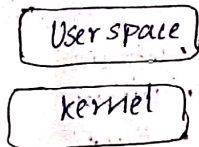
Manages resources of multiple computers to appear as one.

Network OS

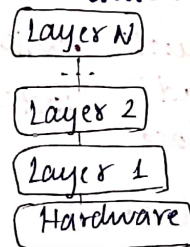
Provides services over a network to other systems.

OS STRUCTURES

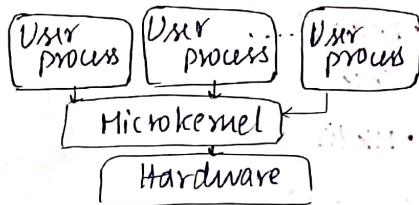
Monolithic Kernel



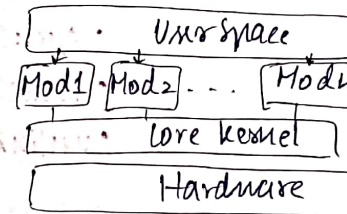
Layered Structure



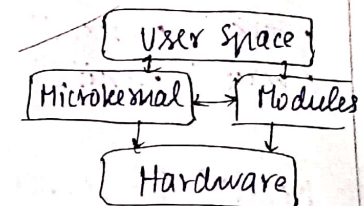
Microkernel



Modular structure



Hybrid Structure



SYSTEM CALLS

Process Control

fork(), exec(),
exit(), wait(),
kill().

File Management

open(), close(),
read(), write(),
lseek(), delete()

Device Management

read(), write(), ioctl(),
open(), close()

Memory Management

mmap(), brk(),
malloc(), free(),
shmget().

Information Management

getpid(), time(),
uname(), getuid(),
getcwd().

OS FUNCTIONS

Process Management

- Process Creation
- CPU Scheduling
- Context switching
- Synchronization

Memory Management

- Memory Allocation
- Virtual memory
- Paging
- Segmentation

File management

- File Organisation
- File storage
- Access Methods
- Directory Mgmt.

Device Management

- Device Allocation
- I/O operations
- Device Drivers
- Buffering

Security & Protection

- Authentication
- Access control
- Data Protection
- Auditing

2.

OPERATING SYSTEMS

OS SERVICES

- CPU scheduling
- Memory Allocation
- Device Handling
- File Management
- Information Mgmt
- Protection & Security

RESOURCE MGMT.

- Manages all system resources
- Allocates and deallocates
- Ensures efficient & fair usage.

SYSTEM CALL INTERFACE

- Provides interface to users
- Bridge b/w user programs and kernel

SYSTEM CALLS (API to kernel)

1. Process System Calls

- `fork()` • `exec()`
- `wait()` • `exit()`

Related OS Service

CPU scheduling &
Process Management

2. File System Calls

- `open()` • `read()`
- `write()` • `close()`
- `unlink()` • `mkdir`

Related OS Service
File Management

3. Device System calls

- `open()` • `write()`
- `read()` • `ioctl()`
- `close()`

Related OS Service
Device handling
(I/O Management)

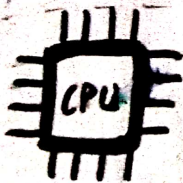
4. Info System Calls

- `getpid()`
- `uname()`
- `time()`
- `getuid()`
- `sysinfo()`

Related OS Service
System Information
Resource Monitoring

LINUX KERNEL

(handles system calls & provides OS services)



CPU
(Processing)



Memory
(Main Memory)



I/O Devices
(Disks, Printers, keyboards, etc.)